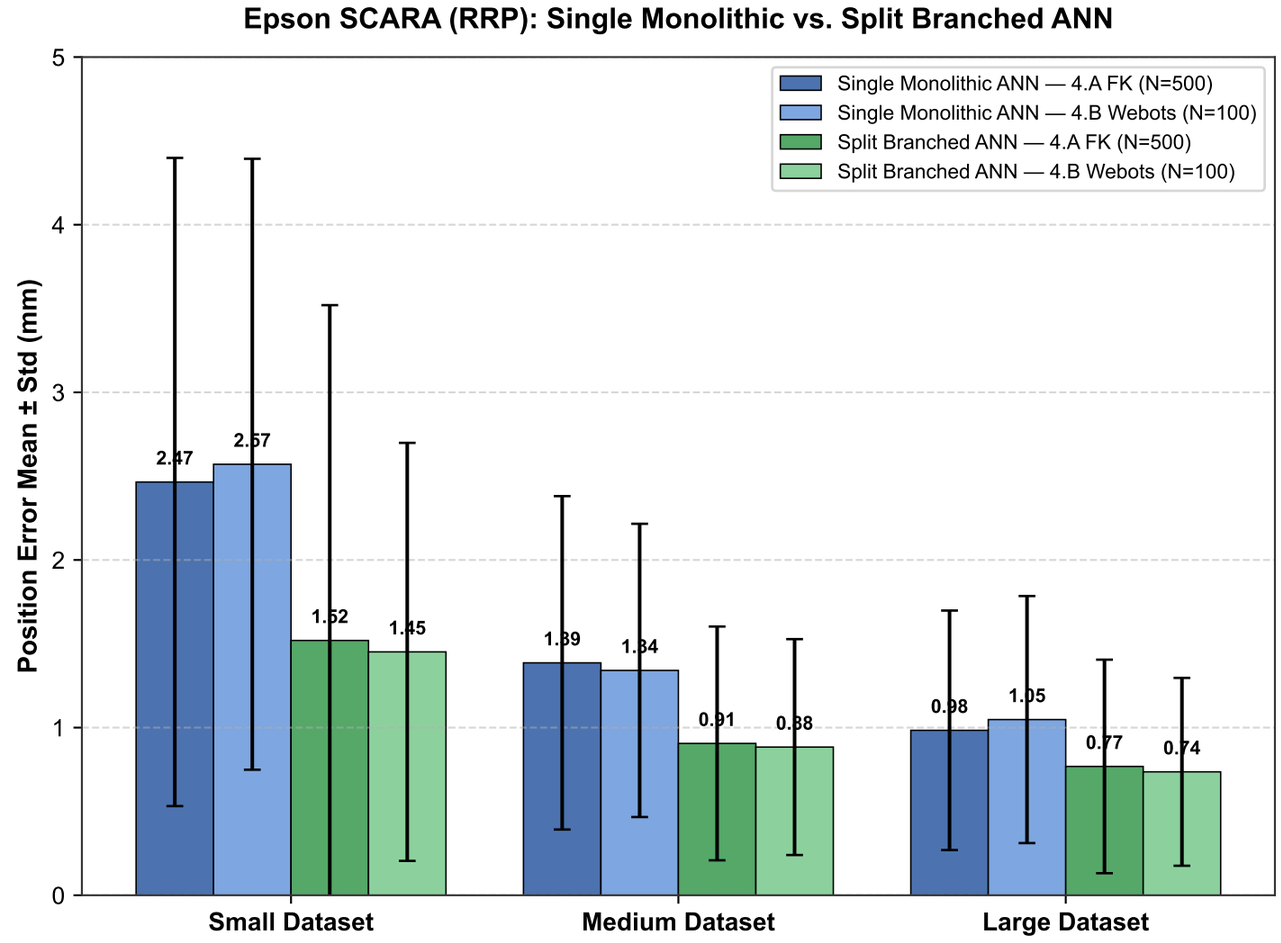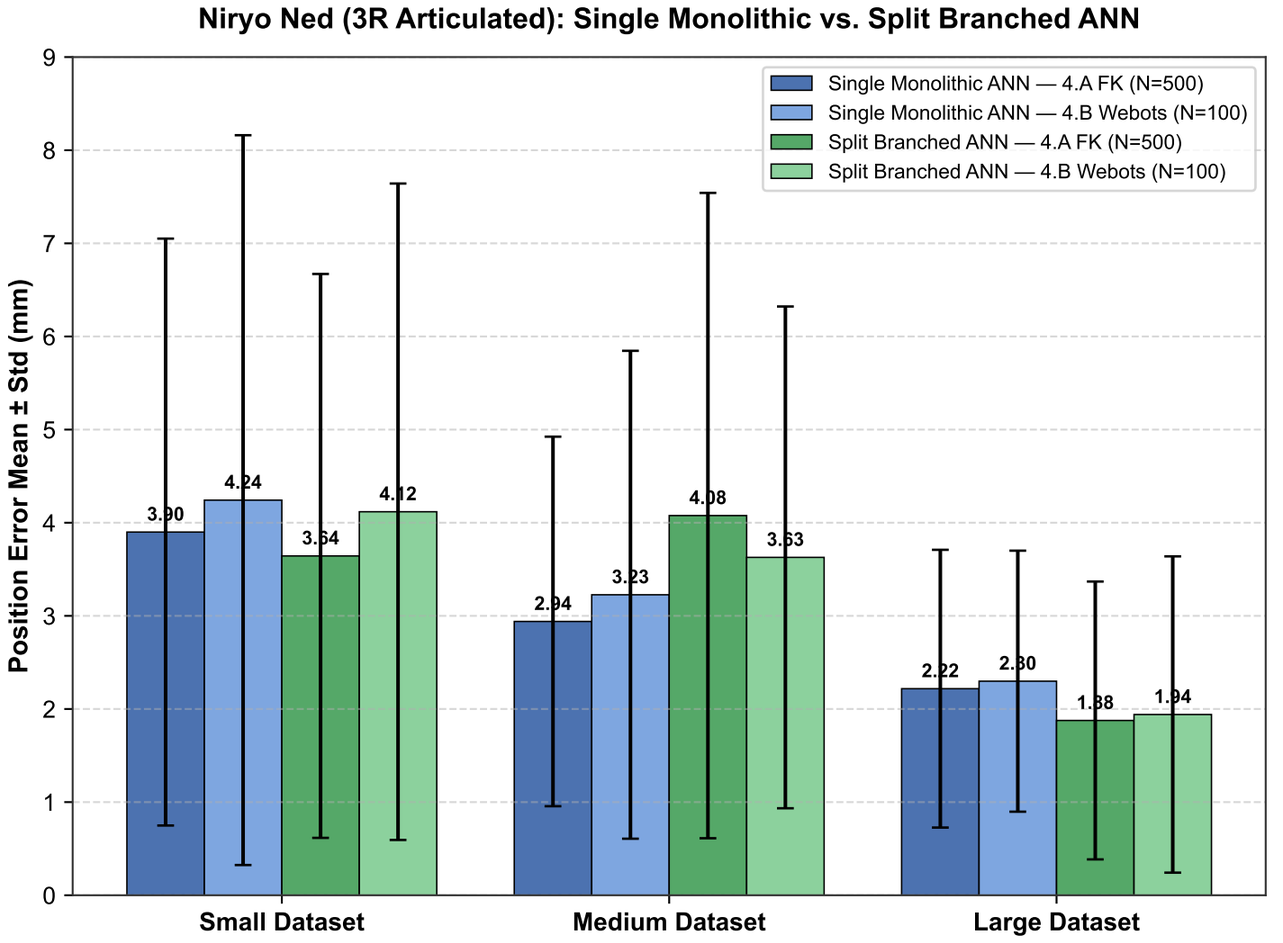Comparative Analysis: Single Monolithic (Input-Gated) vs. Split Branched (Output-Gated) ANN